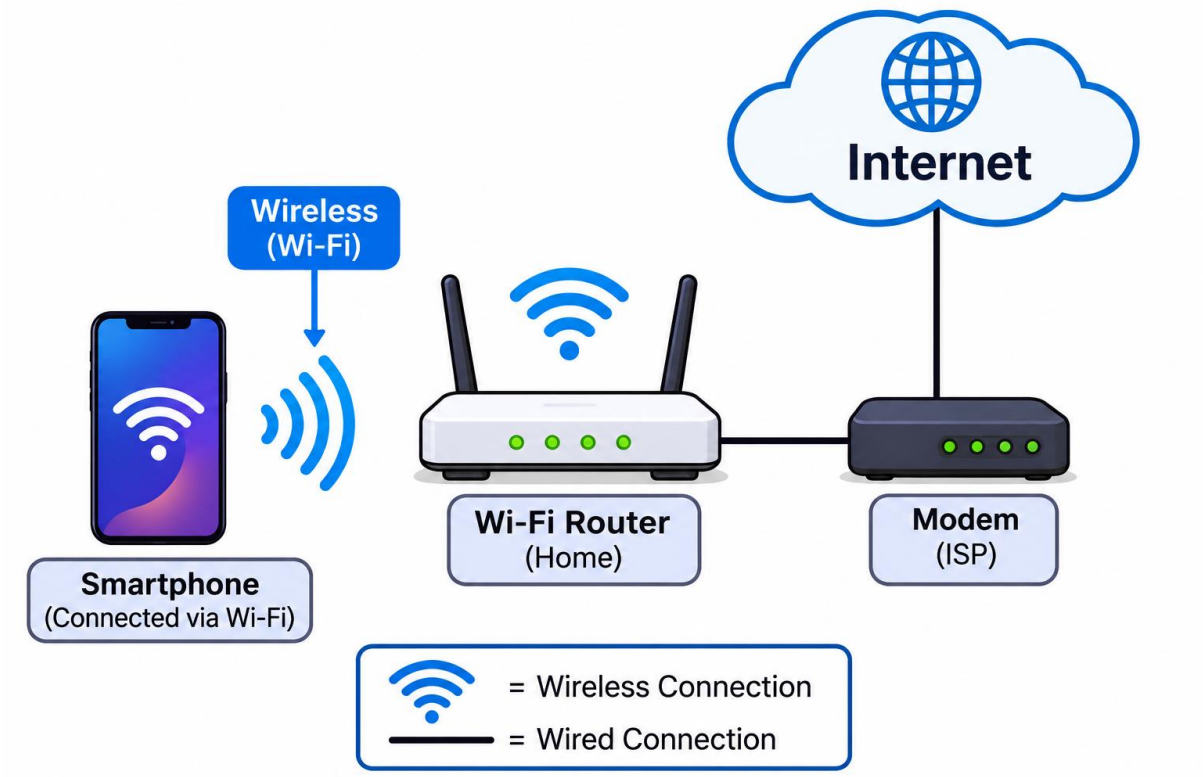
Wireless Networking

Task 1:- List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

Wi-Fi Standard	Frequency Band	Maximum Theoretical Speed
802.11n (Wi-Fi 4)	2.4 GHz and 5 GHz	Up to 600 Mbps
802.11ac (Wi-Fi 5)	5 GHz	Up to 6.9 Gbps
802.11ax (Wi-Fi 6)	2.4 GHz and 5 GHz	Up to 9.6 Gbps

Task 2:- Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.



Task 3:- Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.Consider both signal strength and security when choosing.

Wi-Fi Network Scanning Practical

Step 1: Scan Wi-Fi networks on Windows

Open CMD and run:

```
netsh wlan show networks mode=bssid
```

Note the SSID, Signal, and Authentication type for at least 3 networks.

Step 2: Observation Table (Sample)

SSID	Signal Strength	Security Type
Home_WiFi	90%	WPA2-Personal
JioFiber_5G	75%	WPA3-Personal
Guest_WiFi	45%	Open (No security)

Task 4:- Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

SSID Hiding in Wi-Fi Networks

SSID hiding means disabling the broadcast of a Wi-Fi network's name, so it does not normally appear in the list of available Wi-Fi networks.

- Advantage: It makes the Wi-Fi network less visible to casual users.
- Disadvantage: It does not provide strong security because the network can still be discovered using scanning tools.